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HybridMimic：混合 RL-质心控制用于人形机器人运动模仿

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摘要—运动模仿，即鼓励控制策略模仿人类运动，通过强化学习 ***(RL)*** 促进人形机器人的复杂任务学习。尽管标准 ***RL*** 框架在运动敏捷性方面表现出色，但在部署过程中通常绕过对机器人动力学的显式推理，这种设计选择在机器人遇到分布外环境时可能导致物理上不可行的指令。通过整合基于模型的原理，混合方法可以提高性能；然而，现有方法通常依赖预定义的接触时间，限制了其通用性。本文介绍了 ***HybridMimic***，这是一个框架，其中学习到的策略通过预测连续接触状态和期望质心速度动态地调节基于质心的控制器。这种架构利用质心动力学的物理基础来生成即使在领域偏移下仍保持可行的前馈力矩。使用物理信息奖励，策略被训练以通过输出精确的控制目标和参考力矩来高效利用质心控制器的优化。通过在 ***Booster T1*** 人形机器人上的硬件实验，***HybridMimic*** 与最先进的 ***RL*** 基线相比，将平均基座位置跟踪误差降低了 ***13%***，展示了动态感知部署的鲁棒性。

一、引言

在用于人形机器人的强化学习（RL）的背景下，运动模仿，即通过跟踪奖励来鼓励控制策略模仿人类运动，是一种在实现移动和移动操作任务的同时提高运动自然性和样本效率的技术。运动模仿使人形机器人能够自然地行走、奔跑和跳舞 [1], [2] ，并执行移动操作任务，例如拾取箱子 [3]。随着这些成功，利用运动模仿来构建能够实现各种复杂任务的具身智能，在各个应用中具有广阔前景。这项工作的主要目标是为人形机器人引入HybridMimic，这是一种混合控制架构，将RL与基于模型的控制器集成，使机器人能够学习多样化和物理上合理的运动。该框架解决了在部署标准RL时确保物理可行性的重大挑战，同时消除了传统基于模型控制器对手工制作、预定义接触计划的依赖。

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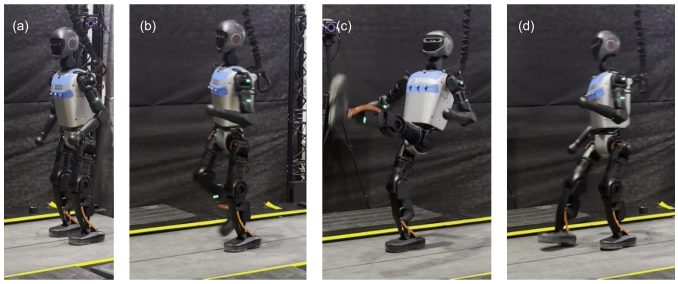

图1：使用所提出的 HybridMimic 控制器执行踢球任务的 Booster T1 的快照。(a) 机器人开始站立。(b) 机器人向前行走。(c) 机器人用左脚踢球。(d) 机器人从踢球中恢复并稳定，向后迈步。更多动作可参考补充视频。https://youtu.be/1d5vkqNtCOY

A. 相关工作

1) 标准RL：RL策略通常在基于物理的模拟器中进行训练，以确保物理可行性，并采用领域随机化 [4], [5] 以确保对各种干扰 [6]的鲁棒性。然而，标准RL框架，特别是那些使用比例-微分（PD）控制器以实现关节级扭矩生成的框架，在部署时不会明确推理机器人动力学。因此，当策略的观测值偏离训练分布时，策略网络缺乏生成物理可行命令的机制，导致仿真到现实的性能下降。

2) 基于分析模型的控制：尽管这些方法有理论基础 [7] 并且在各种环境中取得了令人印象深刻的成功，使机器人能够进行动态移动 [8]–[12],，但基于模型的控制器在需要实现非结构化环境中敏捷和鲁棒移动的任务时面临重大挑战，这是因为相关机器人动力学具有高维、混合、非线性的特性。经典的基于倒立摆的方法 [13]–[17] 通常依赖于简化假设，例如质心（CoM）高度恒定，这些假设在自然的人类类运动（如蹲下或跳跃）期间经常被违反。即使更复杂的质心模型 [18] 通常也需要预定义的刚性接触计划才能保持计算上的可行性。这些手工制作的启发式方法难以推广到“自然场景”中的运动，其中接触时间是非周期的或难以预先指定。此外，这些控制器的性能严格受到底层降阶模型精度的限制；它们缺乏从经验中学习或适应未建模环境细微差别的内在能力。这些局限性要求一种基于学习的方法，该方法可以在仍然尊重基本物理约束的同时发现最优控制策略。

这些控制器的性能严格受到底层降阶模型精度的限制；它们缺乏从经验中学习或适应未建模环境细微差别的内在能力。这些局限性要求一种基于学习的方法，该方法可以在仍然尊重基本物理约束的同时发现最优控制策略。

3) 混合方法：将基于模型的控制器组件或原理集成到强化学习框架中的混合方法已被研究以提升性能。为提高训练效率， [19], [20] 采用基于模型的输出作为参考信号来指导策略训练。同时， [21]–[25]在其强化学习框架中采用基于模型的全身控制器作为力矩生成器。这些框架促进了精确跟踪、 [22], 更高的步行速度 [23],四足跳跃中的敏捷性提升 [24],以及对外部干扰的更好抑制 [25],, 表明包含基于模型的力矩控制器可以显著提升运动性能。

虽然前景广阔，但现有的混合方法依赖于简化假设，这限制了它们在多样化运动模仿中的适用性。基于倒立摆的方法 [23], [25]假设质心高度恒定，这一近似在类人运动中通常不成立。另一种常见方法，质心方法 [22], [24], 通常需要预定义的足步时间。

B. 贡献

在本文中，我们提出了 HybridMimic，一种结合强化学习（RL）与质心模型控制器的混合控制架构，通过质心动力学生成可行的扭矩指令。我们的 HybridMimic 公式通过策略观测实现接触状态估计，无需手工设计的接触计划 [22], [24]。所提出的控制器提供了一个通用框架，通过训练其模仿人类参考运动 [1], [2] （见图1）来学习多样化的运动。这项工作的具体贡献如下：

(a) 无需接触计划的公式：与现有的混合质心方法 [22], [24]，不同，HybridMimic 基于观测估计连续接触状态，无需预定义的接触计划，并提供平滑的接触过渡。此外，HybridMimic 通过考虑策略网络输出的参考扭矩来优化地面反作用力。这使得多样化的地面反作用力曲线自然涌现，使其非常适合运动模仿中固有的复杂接触过渡。

(b) 基于物理的奖励：我们引入了新的奖励项，这些奖励项是质心控制器输入和输出的函数，例如指令加速度、估计的接触状态和地面反作用力。这些奖励鼓励策略对质心控制器进行一致且基于物理的使用，从而产生一个可解释且透明的基于模型的控制器，可以简化物理部署。

(c) 真实世界部署：通过广泛的仿真到仿真和仿真到现实的实验，我们证明了 HybridMimic 成功学习执行多种人类动作，同时保持与最先进基线 [1] 相当的性能。我们进一步展示了基于模型控制的优势，通过展示机器人在运动过程中机器人基座位置跟踪的仿真到现实差距减小，与基线相比，基座位置误差降低了 13%。

II. 前提

我们首先回顾了机器人动力学模型以及 HybridMimic 相关的命名。我们关注全阶动力学以关联前馈扭矩和地面反作用力，以及质心动力学以关联地面反作用力和质心加速度。这些模型通过基于物理的模型将高层质心加速度指令桥接到低层关节扭矩。

A. 全阶动力学

使用多刚体动力学，人形机器人的动力学方程可以表示为

$$M(\tilde{q})\ddot{\tilde{q}} + H(\tilde{q}, \dot{\tilde{q}}) = Bu + \sum_{i \in \mathcal{E}} J_i^\top F_i \tag{1}$$

其中 $\tilde{q} \in \mathbb{R}^{6+n}$ 是浮基人形机器人的广义坐标， n 是关节总数，$M(\tilde{q}) \in \mathbb{R}^{(6+n)\times(6+n)}$ 是惯性矩阵， $H(\tilde{q}, \dot{\tilde{q}}) \in \mathbb{R}^{6+n}$ 是科里奥利、离心和重力项的总和，$u \in \mathbb{R}^n$ 是关节电机扭矩， $B \in \mathbb{R}^{(n+6)\times n}$ 是输入矩阵，而 $F_i \in \mathbb{R}^6$ 和 $J_i \in \mathbb{R}^{6\times(n+6)}$ 是索引为 $i \in \mathcal{E}$ 的地面反作用力偶和相应的雅可比矩阵。集合 $\mathcal{E}$ 表示机器人上所有可能的接触表面。在本工作中，考虑了一个有限的集合 $\mathcal{E} = \{$ 左脚、右脚$\}$ 。然而，给定可行的运动片段，所提出的框架可以 readily 扩展以考虑更多可能的接触表面，例如手、膝盖和肘部。

B. 质心动力学

为实现在高频控制并避免在强化学习训练过程中进行复杂计算，我们使用单个刚体（SRB）模型来捕捉人形机器人的质心动力学。SRB模型假设机器人的质量集中在质心（CoM）处，转动惯量保持不变，并且可以通过地面反作用力矩F_i 以及所有 $i \in \mathcal{E}$ [18]的力臂 $r_i \in \mathbb{R}^3$ 来控制运动。质心的线性加速度 $\ddot{x}_{\text{lin}} \in \mathbb{R}^3$ 和角加速度 $\ddot{x}_{\text{ang}} \in \mathbb{R}^3$ 可以表示为

$$m\ddot{x}_{lin} = mg + \sum_{i \in \mathcal{E}} F_{i,\text{lin}} \tag{2}$$

$$I_c\ddot{x}_{\text{ang}} = \sum_{i \in \mathcal{E}} F_{i,\text{ang}} + [r_i]_\times F_{i,\text{lin}} \tag{3}$$

其中 $m \in \mathbb{R}$ 是机器人的总质量， $g \in \mathbb{R}^3$ 是重力加速度矢量，$F_{i、\text{lin}} \in \mathbb{R}^3$ 是线性部分

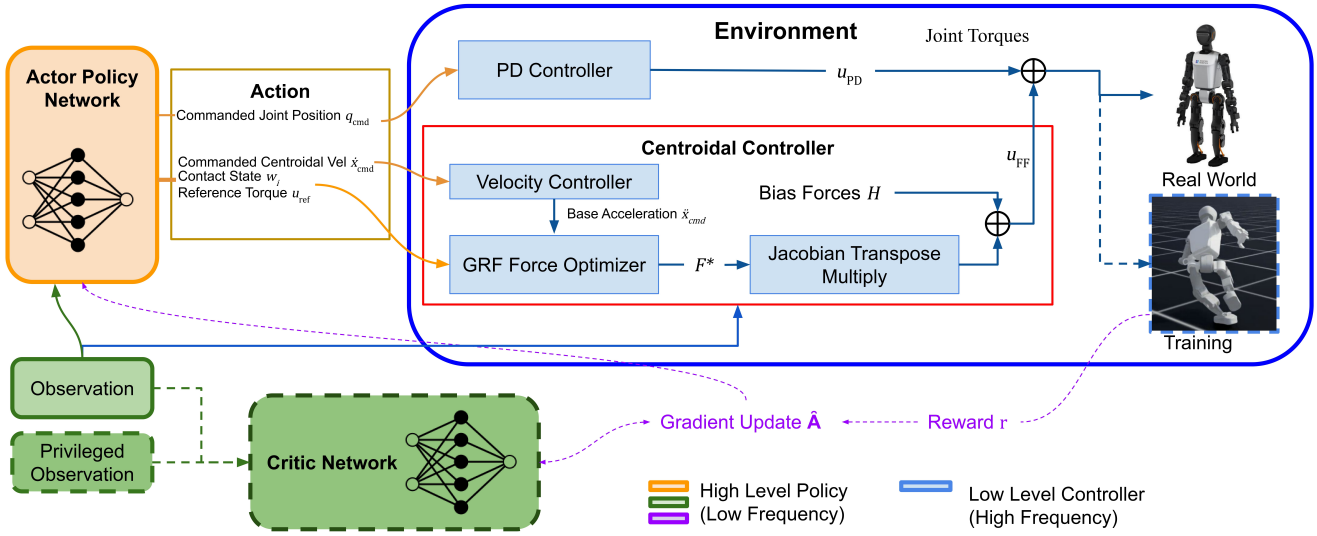

图2：展示HybridMimic控制器在训练和部署中的公式图。虚线项表示训练中特有的元素，在部署时未使用。

F_i 的旋转部分，$I_c \in \mathbb{R}^{3\times3}$ 是机器人关于质心的转动惯量矩阵，$F_{i,\text{ang}} \in \mathbb{R}^3$ 是 F_i 的角速度部分，而 $[r_i]_\times$ 是 r_i 的反对称矩阵。将上述两个方程堆叠起来得到：

$$\begin{bmatrix}\ddot{x}_{\text{lin}}\\ \ddot{x}_{\text{ang}}\end{bmatrix} = \hat{g} + \sum_{i\in\mathcal{E}} A_i F_i \tag{4}$$

其中 $\hat{g} = [g^\top, 0_{1\times3}]^\top$ $F_i = [F_{i,\text{lin}}^\top \quad F_{i,\text{ang}}^\top]^\top$ $A_i = \begin{bmatrix} \frac{1}{m} I_3 & 0 \\ I_c^{-1}[r_i]_\times & I_c^{-1} \end{bmatrix}$，lin，ang，和 。该表达式可简化为

$$\ddot{x} = \hat{g} + AF \tag{5}$$

其中 $\ddot{x} = [\ddot{x}_{\text{lin}}^\top, \ddot{x}_{\text{ang}}^\top]^\top$，$A$ 水平堆叠所有 $A_i\ \forall i \in \mathcal{E}$，而 F 垂直堆叠所有 $F_i\ \forall i \in \mathcal{E}$。

III. 方法论

为了模仿从人类演示中重定向的运动参考 $\tilde{q}_{\text{ref}}(t)$，控制器需要生成电机扭矩$u(t)$，以最小化运动参考$\tilde{q}_{\text{ref}}(t)$与机器人实际运动 $\tilde{q}(t)$ 之间的差异。为简化起见，本文其余部分省略了时间显式依赖。受模型控制 [27], [28], 启发，我们的 HybridMimic 控制器（图2）将电机扭矩计算为前馈扭矩 u_{FF} 和 PD 扭矩 u_{PD} 的总和：

$$u = u_{\text{FF}} + u_{\text{PD}} \tag{6}$$

与标准RL [1], [2], 类似，PD项被设计为

$$u_{\text{PD}} = K_P(q_{\text{cmd}} - q) + K_D\dot{q} \tag{7}$$

K_P 和 K_D 分别是电机的刚度和阻尼矩阵，$q \in \mathbb{R}^n$ 是实际关节位置，$q_{\text{cmd}} \in \mathbb{R}^n$是由策略网络生成的指令关节位置。仅使用PD项已被证明在移动任务中具有经验上的有效性，并且可以隐式地处理前馈和反馈项 K_P。

然而，PD公式将所有控制命令编码到一个由策略控制的单一通道 q_{cmd} 中。Xiong等人 [30]指出，标准PD公式在没有前馈速度指令的情况下可能会造成延迟。

因此，除了PD项外，还使用了一个质心控制器来生成用于HybridMimic的前馈扭矩 u_{FF}。

A. 前馈扭矩生成

为了补偿接触力矩和偏置力 H的影响，前馈扭矩由以下方式生成：

$$u_{\text{FF}} = -B^\dagger \sum_{i\in\mathcal{E}} J_i^\top F_i^* + B^\dagger H \tag{8}$$

其中 $B^\dagger$ 是 B 的伪逆，F_i^*是接触 $i \in \mathcal{E}$的可行地面反作用力估计。

通过求解以下约束二次规划(QP)问题来获得接触力矩 $F_i^*\ \forall i \in \mathcal{E}$的可行估计，以尊重质心动力学和估计的接触状态：

$$\begin{aligned} \min_F \quad & L(F) \\ \text{subject to} \quad & AF = \ddot{x}_{\text{cmd}} - \hat{g} \end{aligned} \tag{9}$$

命令质心加速度 $\ddot{x}_{\text{cmd}}$ 的确定位置

$$\ddot{x}_{\text{cmd}} = K_{\text{vel}}(\dot{x}_{\text{cmd}} - \dot{x}) \tag{10}$$

这里，$K_{\text{vel}} \in \mathbb{R}^{6\times6}$ 是质心速度跟踪的增益，$\dot{x}_{\text{cmd}} = [\dot{x}_{\text{cmd,lin}}^\top, \dot{x}_{\text{cmd,ang}}^\top]^\top \in \mathbb{R}^6$ 是策略网络指令的质心速度输出，$K_{\text{vel}} \in \mathbb{R}^{6\times6}$ 是实际质心速度。成本函数 $\dot{x}_{\text{cmd}} = [\dot{x}_{\text{cmd,lin}}^\top, \dot{x}_{\text{cmd,ang}}^\top]^\top \in \mathbb{R}^6$ () 定义为

表 I：Actor 和 critic 的观测。Actor 可以访问基线观测，而 critic 则同时拥有基线和特权观测。四个策略时间步的观测历史被堆叠在一起。

Term	观测	维度
基线观测		
q_{ref}	参考关节位置	n
$\dot{q}_{\text{ref}}$	参考关节速度	n
$\dot{P}_B$	基座帧线性速度	3
ω_B	基座帧角速度	3
q	当前关节位置	n
$\dot{q}$	当前关节速度	n
优先观察		
p_{ref}	参考基座在世界帧中的位置	3
p	当前世界坐标系中的基座位置	3
$R_{WB,\text{ref}}$	参考基坐标系相对于世界坐标系（前两个的6个元素列于一个 3×3 旋转矩阵）	6
R_{WB}	当前基坐标系方向相对于世界坐标系	6

策略网络输出的速度，$\dot{x}\in\mathbb{R}^6$ 是实际的质心速度。成本函数 $L(F)$ 定义为

$$L(F)=\underbrace{\|u_{\text{ref}}+\sum_{i\in\mathcal{E}}J_i^\top F_i\|^2_{K_\tau\,\text{diag}(\tau_{\text{limit}}^{-2})}}_{\text{Reference torque cost}}+\underbrace{\sum_{i\in\mathcal{E}}\left\{||F_{i,\text{lin}}||^2_{e^{-w_i}K_{\text{lin}}}+||F_{i,\text{ang}}||^2_{e^{-w_i}K_{\text{ang}}}\right\}}_{\text{End-effector contacting cost}} \tag{11}$$

其中 $u_{\text{ref}}\in\mathbb{R}^n$ 是策略生成的参考扭矩，K_τ 是参考扭矩成本的名义权重，并按电机扭矩限制 $\tau_{\text{limit}}\in\mathbb{R}^n$ 进行缩放，其中 τ^{-2} 限制表示 τ_{limit} 的逐元素平方倒数。$w_i\in\mathbb{R}$ 是策略生成的 i^{th} 接触表面在 $\mathcal{E}$ 的接触状态。K_{lin} 和 K_{ang} 是末端执行器接触成本的名义权重，并按接触状态 w_i 进行缩放。较大的 w_i 意味着 i^{th} 表面更有可能与环境接触。我们采用接触状态的连续描述，以确保策略动作到关节扭矩的连续映射，从而避免可能阻碍 RL 训练的扭矩突变。

成本函数可以表示为二次型 $L(F)=\frac{1}{2}F^\top QF+c^\top F$，其中 Q 和 c 的详细定义被省略。这允许 (9) 的最优解以如下形式表示为封闭形式，这便于在训练和实际部署期间进行高效计算：

$$F^*=-Q^{-1}c+Q^{-1}A^\top\left(AQ^{-1}A^\top\right)^{-1}\left(\ddot{x}-\hat{g}+AQ^{-1}c\right). \tag{12}$$

最后，前馈扭矩被限制以尊重电机扭矩限制 τ_{limit}，以确保安全性和可行性。

在我们的HybridMimic框架中，策略网络生成输出以调整参考扭矩 u_{ref}、i^{th} 接触的接触状态w_i 以及指令质心速度$\dot{x}_{\text{cmd}}$。这些输出使策略网络能够利用QP问题的结构和SRB动态模型的结构

表II：奖励项的值。奖励项分为跟踪、正则化和HybridMimic三类。跟踪项直接来自 [1]并调整了参数。HybridMimic项在III-B1、III-B2、III-B3和III-B4节中描述。对于具有灵敏度σ的项，奖励定义为 $r=e^{\sigma^2}$，其中 ϵ 表示期望值与测量值之间的误差。

Type	奖励	权重	敏感度
跟踪	全局锚点位置	0.7	0.3
	全局锚点方向	0.7	0.4
	身体位置	1	0.2
	身体方向	1	0.3
	身体线速度	1	0.5
	身体角速度	1.25	2.09
正则化	L2动作速率	-0.08	-
	非期望接触	-0.5	-
HybridMimic	GRF奖励	0.15	100
	接触状态奖励	0.5	-
	扭矩限制奖励	0.1	-
	质心加速度（线性）	0.1	6
	质心加速度（角）	0.1	12

在不了解接触计划先验知识的情况下生成前馈扭矩。通过使用 w_i缩放末端执行器接触成本权重，该公式仅对具有高接触状态值的接触表面产生显著的地面反作用力。通过参考扭矩成本项，策略网络对 F^*获得更好的控制，尤其是在(9)中的等式约束有无限多解时（例如，在双支撑阶段），并隐式防止摩擦锥和驱动限制的违反。

B. RL策略

为训练策略网络以利用(6)中提出的混合控制器，将RL问题制定为部分可观察马尔可夫决策过程（POMDP），表示为元组 $\langle\mathcal{S},\mathcal{O},\mathcal{A},\mathcal{P},r,\gamma\rangle$。其中，$\mathcal{S}$是状态空间，$\mathcal{O}$ 是观察空间（表I），$\mathcal{A}$ 是动作空间（包括u_{ref}, $w_i\ \forall i\in\mathcal{E}$, $\dot{x}_{\text{cmd}}$和 q_{cmd}），$\mathcal{P}_{S'|S,A}$ 是状态转移概率, $r:\ \mathcal{O}\times\mathcal{A}\to\mathbb{R}$ 是奖励函数（表II），γ 是折扣因子。策略 $\pi_\theta:\ \mathcal{O}\to\mathcal{A}$, 由 θ参数化，被训练以最大化从初始分布 $s_0\sim p_0$中采样的初始状态开始的轨迹上的预期折扣回报，其中在时间步 t处根据观察 o_t 选择动作 a_t：

$$\mathbb{E}_{s_0\sim p_0(\cdot),\ s_{t+1}\sim\mathcal{P}(\cdot|s_t,a_t),\ a_t\sim\pi_\theta(\cdot|o_t)}\left[\sum_{t=0}^{\top}\gamma^\top\,r(s_t,a_t)\right] \tag{13}$$

我们采用近端策略优化 [31] 和非对称演员-评论家 [32] 进行高效学习，为每个运动片段训练一个策略。为了提高鲁棒性，我们使用领域随机化，具体如 Tab.III 所述。

这项工作的一个关键创新是引入了确保质心控制器可行利用的奖励函数。我们通过奖励设计，通过最小化模拟值（如地面反作用力 F_{sim}）与预测值 F^* 之间的误差，来鼓励控制器的可行利用。我们的奖励促进了推导

TABLE III: Settings of domain randomization applied to improve policy robustness. Terms with + have a uniformly distributed random noise added to the true value. Terms with * have a uniformly distributed random factor multiplied to the true value. We add random noise to HybridMimic inputs to prevent the policy from learning behaviors that exploit using the centroidal controller for noise-free observation.

Type	Randomization Term	Range
Observations	Base linear velocity noise (m/s)	+(-0.5, 0.5)
	Base angular velocity noise (rad/s)	+(-0.2, 0.2)
	Joint position noise (rad)	+(-0.02, 0.02)
	Joint velocity noise (rad/s)	+(-0.5, 0.5)
HybridMimic input	Mass randomization M (kg)	*(0.98, 1.02)
	Inertia matrix I_c	*(0.98, 1.02)
	Jacobian J	*(0.98, 1.02)
	End-effector position r (m)	+(-0.02, 0.02)
Environment	Initial joint position (rad)	+(-0.1, 0.1)
	Body CoM position offset (m)	+(-0.025, 0.025)
	Base velocity push (m/s)	+(1.0, 3.0)

of accurate, interpretable outputs, including ground reaction wrenches, contact states, and centroidal acceleration.

The following sections describe the specific rewards implemented for HybridMimic.

1) Ground reaction force (GRF) reward: The policy network should, when utilizing the centroidal controller correctly, generate an estimate of ground reaction wrench F_i^* close to the value $F_{i,\text{sim}}$ from the simulator for the i^{th} contact. Thus, a reward is given for minimizing the error between the two values.

$$r_{\text{GRF}} = \exp\left(-\frac{\sum_{i\in\mathcal{E}} \|F_{i,\text{sim}} - F_i^*\|^2}{\sigma_{grf}^2}\right) \tag{14}$$

The sensitivity constant $\sigma_{grf} \in \mathbb{R}$ can be chosen based on the magnitude of the expected contacting wrench of the humanoid robot.

2) Contact state reward: The contact state w_i generated by the policy network should match the contact state $w_{i,\text{sim}}$ obtained through the simulator. A penalty is given for the predicted contact state of the policy network deviating from the true contact state

$$r_{\text{contact}} = -\sum_{i\in\mathcal{E}} (\text{sigmoid}(w_{i,\text{sim}}) - \text{sigmoid}(w_i))^2 \tag{15}$$

3) Torque limit reward: For computational efficiency, the centroidal controller does not consider torque limits. The policy network is intended to implicitly learn to utilize u_{ref} to avoid exceeding torque limits. A torque limit penalty is applied when u_{FF} exceeds a soft torque limit.

$$r_{\tau_{\text{limit}}} = -\sum_{j=1}^{n} \max(0, u_{\text{FF},j} - \alpha\tau_{\text{limit},j}) \tag{16}$$

where $\alpha \approx 0.9$ and the subscript $[\cdot]_j$ indicates the j^{th} joint.

4) Centroidal acceleration reward: The robot's acceleration in simulation should match the commanded acceleration computed by equation (10). To encourage this behavior, a centroidal acceleration reward is implemented for minimizing the error between the simulated acceleration $\ddot{x}_{\text{sim}}$ and the commanded centroidal acceleration $\ddot{x}_{\text{cmd}}$.

$$r_{acc} = \exp\left(-\frac{\|\ddot{x}_{\text{cmd}} - \ddot{x}_{\text{sim}}\|^2}{\sigma_{acc}^2}\right) \tag{17}$$

where $\sigma_{acc} \in \mathbb{R}$ is the sensitivity constant detailed in Table II.

IV. Results and Analysis

To validate the effectiveness of the proposed HybridMimic framework to mimic human motion, we conduct an extensive study of both sim-to-sim and sim-to-real transfer. In the study, several variations of the controller are compared to show that the proposed design is crucial to achieving the improved motion mimicking performance. The controllers include:

(i) **BeyondMimic** [1]: This produces an RL policy using only a PD controller (7) as the torque generator, and serves as our baseline to highlight the performance gains introduced by the centroidal controller.

(ii) **HybridMimic**: The proposed controller from Sec. III.

(iii) **HybridMimic with fixed contact schedule (FCS) (termed as "HybridMimic+FCS")**: This is HybridMimic but with the contact state w_i derived from the reference motion. We set $w_i = 1$ if the distance between the i^{th} contact surface and the environment is below a threshold and $w_i = 0$ otherwise. This ablation is used to demonstrate that the policy-estimated contact states w_i are necessary for robust motion mimicking.

(iv) **HybridMimic+FCS without Reference Torque Cost (RTC) (termed as "HybridMimic+FCS-RTC")**: This is the HybridMimic+FCS formulation with the reference torque cost removed from the cost function $L(F)$. As our most complete ablation, this setup closely mirrors previous hybrid centroidal controllers [22], [24]. We use it to show that policy-generated contact states w_i and reference torques u_{ref} are essential for robust performance.

To evaluate the performance of a controller, we compare the difference between the trajectory rollout in the training environment q_{train} (which is a good approximation of q_{ref} after sufficient training) and the rollout during evaluation q_{eval} either in the MuJoCo simulation or in the real experiment. We use q_{train} as a baseline instead of the motion reference q_{ref} since the latter may not be dynamically feasible to be tracked by the robot due to retargeting issues [26]. The difference between q_{eval} and q_{train} also characterizes the capability of the controller under domain shift caused by sim-to-sim or sim-to-real transfer. In this study, the following performance metrics are used:

(a) **Base final position error (m)**: The L2 norm of the base position error at the end of the motion clip.

(b) **Base mean position error (m)**: The averaged L2 norm of the base position error across the entire motion clip.

(c) **Base mean orientation error (rad)**: The averaged L2 norm of orientation error across the entire motion clip.

(d) **Base velocity root-mean-square (RMS) error (m/s)**: The RMS value of the L2 norm of linear velocity error for the entire motion clip.

TABLE IV: Training and HybridMimic parameters.

Training Parameters			
Term	Value	Term	Value
# of environments	8192	# of iterations	15000
Episode length	10s	Learning rate	10^{-3}
Desired KL	0.01	Value loss coeff.	1.0
Entropy coeff.	0.005	Clip ratio	0.2
Actor size	[1024,512,256,256]	γ (Discount factor)	0.99
Critic size	[1024,512,256,256]	λ (GAE decay)	0.95
HybridMimic Parameters			
Term	Value	Term	Value
$K_{\text{vel,lin}}$	$5I_3$	$K_{\text{vel,ang}}$	$10I_3$
K_τ	$100I_n$	K_{lin}	$10^{-3}I_3$
K_{ang}	$0.02I_3$		

(e) **Base angular velocity RMS Error (rad/s)**: The RMS value of the L2 norm of angular velocity error for the entire motion clip.

A. Training

We implemented HybridMimic in IsaacLab [33] and trained our policies on a Booster T1 humanoid robot for various motion clips. The parameters used in the training environment are described in Tab. IV. The policy network is operated at 50 Hz and the centroidal controller runs at 500 Hz. The stiffness matrix K_P for HybridMimic is 0.75 times that suggested in [1]. When training the policy on a walking task using a partial A100 GPU instance provided by [34], we measured the wall clock time to be 15.4 hours for BeyondMimic and 22.3 hours for HybridMimic, representing a computational overhead of 31% in training time.

B. Sim-to-Sim Results

We train all four policies on three 10-s motion clips:

(1) **Walking to kneeling**: Robot walks forward then high kneels on the right leg [35].
(2) **Running in Circle**: Robot runs in a circle [36].
(3) **Jumping**: Robot jumps forward using both legs [37].

We evaluated the trained policies in the IsaacLab training environment without domain randomization to obtain datasets of 256 trained trajectories q_{train}, with a large count to mitigate the nondeterministic effects of the IsaacLab environments. We then evaluate the policies in MuJoCo to obtain q_{eval}. The sim-to-sim tracking results are shown in Tab. V, and the video can be found in the supplementary video.

For all three motion clips, HybridMimic and BeyondMimic are able to mimic the reference motion, while HybridMimic demonstrates lower tracking error in linear position, linear velocity, and angular velocity, showing the proposed framework improves the motion mimicking performance through the feedforward torque. By comparing the HybridMimic with its variants, it can be found that the HybridMimic+FCS and HybridMimic+FCS-RTC have higher tracking error than full HybridMimic for the first two motion clips, where the contact timing is more complex than the jumping motion. The strong performance of HybridMimic+FCS-RTC in the jumping task likely stems from the motion's simple contact schedule, which allows for easier learning due to the lack of redundant action space in full HybridMimic. However, the fragility of fixed contact schedules becomes evident in running, where complex contact patterns and disturbances require adaptable scheduling.

C. Hardware Experiments

We train the HybridMimic and BeyondMimic policies on a series of motion clips involving dynamic locomotion tasks:

(1) **Forward walking**: Starting from standing, the robot walks forward [38]. The task has duration $7s$.
(2) **Side-stepping**: Sidestep to the right [36]. The task has duration $6.5s$.
(3) **Backwards stepping**: Stepping backwards [36]. The task has duration $6s$.
(4) **Kicking**: Robot steps forward then kicks with the left leg before stepping backwards to original location [38]. The task has duration $8s$.

The trained policies are deployed on the real robot. VICON motion-capture cameras are used both to capture tracking data and to provide linear velocity information for policy observation at 100 Hz. On the real robot, the centroidal controller was run at 350 Hz due to hardware limitations. Each policy was executed to obtain three successful trials per motion clip. HybridMimic performs complex dynamic tasks in the real world (Fig. 1, supplementary video), with tracking statistics shown in Tab. V.

As shown in Tab. V, HybridMimic achieves an about 13% reduction in base position tracking error across all four real-world motions compared to the BeyondMimic baseline. This demonstrates that the proposed HybridMimic improves the sim-to-real tracking performance compared to a state-of-the-art baseline. For the sidestepping motion, Fig. 3 shows the robot's base position trajectory in the direction of travel. As shown in Fig. 3, the proposed HybridMimic performs a smoother motion and stays closer to the trajectory shown in training environments, while the motion controlled by BeyondMimic is more jittery and has a larger base position tracking error and can be illustrated in the experiment video.

During sim-to-real deployment, we noted that tuning HybridMimic parameters was comparatively straightforward due to the transparent and explainable nature of the proposed design. One example involves adjusting the gain K_{vel} for centroidal velocity tracking to balance the sensor noise level and the velocity tracking performance.

D. Results Analysis

Alongside demonstrating the reduced tracking error, another feature of HybridMimic is the transparent and interpretable behavior of the centroidal controller. In this section, we discuss the ground reaction force solved from the QP problem (9) and the feedforward torque generated by HybridMimic and its variant during the walking then kneeling motion task in a sim-to-sim study.

The estimated ground reaction forces of HybridMimic and HybridMimic+FCS are shown in Fig. 4. HybridMimic learns to accurately estimate ground reaction forces, with a small error between the simulated and estimated ground reaction

TABLE V: Sim-to-sim and sim-to-real tracking errors across motion clips and controller variants. The bolded figures represent the **lowest** tracking error in the category. The underlined figures represent the second best in the category. The text with background color is the proposed method.

Task	Model	Base Final Position Error (m)	Base Mean Position Error (m)	Base Mean Orientation Error (rad)	Base Velocity Error (m/s)	Base Angular Velocity Error (rad/s)
Sim-to-Sim Results						
Walking to Kneeling	BeyondMimic	0.4739 ± 0.031	0.3344 ± 0.021	0.1577 ± 0.012	0.1331 ± 0.004	0.2881 ± 0.007
	HybridMimic+FCS	0.6824 ± 0.021	0.4768 ± 0.014	0.2356 ± 0.007	0.1291 ± 0.002	0.3021 ± 0.004
	HybridMimic+FCS-RTC	0.3641 ± 0.015	0.2633 ± 0.013	**0.1513 ± 0.006**	0.1202 ± 0.004	0.2679 ± 0.007
	HybridMimic	**0.3455 ± 0.011**	**0.2366 ± 0.008**	0.1576 ± 0.003	**0.1105 ± 0.001**	**0.2285 ± 0.002**
Running in Circle	BeyondMimic	0.3435 ± 0.008	0.2653 ± 0.007	0.1597 ± 0.005	0.1985 ± 0.002	0.3961 ± 0.002
	HybridMimic+FCS	0.5626 ± 0.042	0.2930 ± 0.031	0.1757 ± 0.013	0.2066 ± 0.003	0.5064 ± 0.004
	HybridMimic+FCS-RTC	1.8963 ± 0.009	0.7606 ± 0.009	0.4071 ± 0.005	0.4894 ± 0.000	0.8870 ± 0.001
	HybridMimic	**0.3289 ± 0.012**	**0.2622 ± 0.010**	**0.1413 ± 0.009**	**0.1497 ± 0.001**	**0.3711 ± 0.004**
Jumping	BeyondMimic	0.4670 ± 0.018	0.2794 ± 0.010	0.2827 ± 0.006	0.5346 ± 0.001	1.4305 ± 0.002
	HybridMimic+FCS	0.3771 ± 0.012	0.2225 ± 0.007	0.1518 ± 0.004	0.2168 ± 0.001	0.5657 ± 0.002
	HybridMimic+FCS-RTC	**0.3571 ± 0.009**	**0.1929 ± 0.006**	**0.1253 ± 0.004**	**0.1957 ± 0.000**	**0.4566 ± 0.002**
	HybridMimic	0.3926 ± 0.017	0.2217 ± 0.009	0.1521 ± 0.005	0.2306 ± 0.002	0.5548 ± 0.003
Sim-to-Real Results						
Sidestepping	BeyondMimic	0.0942 ± 0.042	0.0471 ± 0.003	**0.0614 ± 0.004**	0.2767 ± 0.092	**0.7708 ± 0.006**
	HybridMimic	**0.0694 ± 0.008**	**0.0401 ± 0.002**	0.0962 ± 0.001	**0.2676 ± 0.078**	0.7810 ± 0.001
Backwards Stepping	BeyondMimic	0.1849 ± 0.002	0.0500 ± 0.001	0.0912 ± 0.002	0.1994 ± 0.001	0.9044 ± 0.003
	HybridMimic	**0.1079 ± 0.011**	**0.0405 ± 0.002**	**0.0834 ± 0.004**	**0.1947 ± 0.005**	**0.8978 ± 0.003**
Forwards Walking	BeyondMimic	0.2275 ± 0.067	0.0592 ± 0.006	**0.0808 ± 0.011**	0.3293 ± 0.036	0.8762 ± 0.078
	HybridMimic	**0.1377 ± 0.025**	**0.0496 ± 0.001**	0.0912 ± 0.006	**0.1856 ± 0.004**	**0.7581 ± 0.001**
Kicking	BeyondMimic	0.1564 ± 0.051	0.0580 ± 0.001	**0.1338 ± 0.007**	0.9608 ± 0.016	0.6388 ± 0.014
	HybridMimic	**0.1131 ± 0.009**	**0.0567 ± 0.002**	0.1826 ± 0.005	**0.9402 ± 0.008**	**0.6360 ± 0.006**

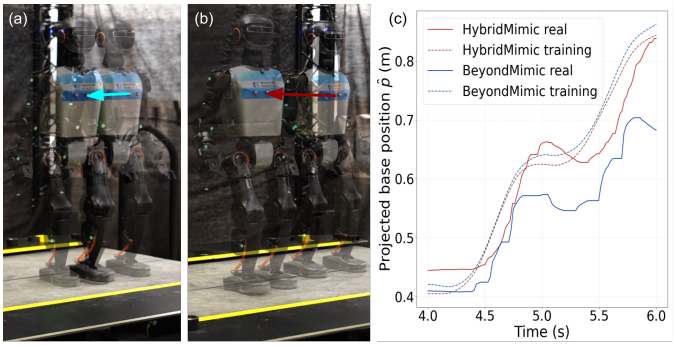

Fig. 3: The left two images illustrate the robot displacement after two sidesteps for (a) BeyondMimic and (b) HybridMimic controllers. (c) The projected base position $\hat{p}$ is the base position projected onto the line from the starting to ending position in the reference motion clip. While the real HybridMimic trajectory oscillates around the trained motion, the real BeyondMimic trajectory consistently undershoots it and induces trunk wobbling as shown in the experiment video.

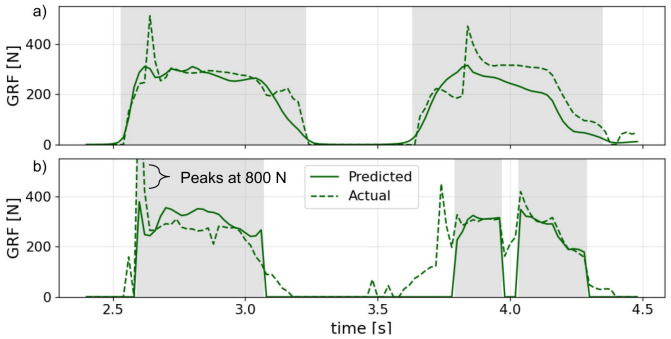

Fig. 4: The vertical ground reaction force of left foot during walking in simulation for (a) HybridMimic and (b) HybridMimic+FCS. The highlighted regions indicate periods of time during which the contact state indicates the left foot is contacting the ground. For HybridMimic, the highlighting occurs when $w_{\text{left foot}} > 0$ and for HybridMimic+FCS , the highlighting occurs when the reference motion has a foot height below 0.007m.

force. Visual illustration can be found in the supplementary video. Additionally, we compare it to HybridMimic+FCS and show that learning contact states is crucial because errors in the contact schedule derived from the motion reference lead to brittle and inaccurate ground reaction force computation. This downgrades the tracking performance through inaccurate feedforward torque generation in (8) and requires the policy to generate additional PD torque to compensate for it.

Figure 5 shows the feedforward torque u_{FF}, reference torque $u$ref, and simulated torque for the left and right knee joints under HybridMimic. When a foot contacts the ground, the actual and feedforward torques are similar, indicating that most motor torque is generated by the centroidal-controller-driven feedforward term. During swing phase, the feedforward torque remains small but nonzero due to the H term in (8). The reference torque roughly follows the feedforward torque but is not expected to match it, as it is an additional control channel for the policy.

V. Conclusion and Future Work

This paper has introduced HybridMimic, a hybrid framework integrating centroidal-model-based control with reinforcement learning (RL) to track diverse, dynamic motion ref-

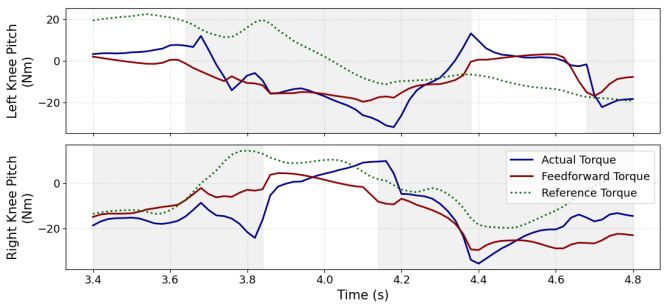

Fig. 5: Feedforward, actual, and reference torque during walking for the left and right knee pitch joints. The highlighted region corresponds to contact on the corresponding leg determined by when $w_i > 0$.

erences. Through physics-informed reward design, the policy network learns to modulate commanded centroidal velocities and continuous contact states. This allows the centroidal controller to generate feedforward torques that effectively counteract ground reaction wrenches. Experimental results on the Booster T1 humanoid demonstrated that HybridMimic achieved a 13% reduction in base position tracking error compared to a state-of-the-art RL baseline during sim-to-real transfer. The synergy between centroidal control and RL facilitates deterministic parameter tuning, such as adjusting velocity tracking gains, which streamlines the deployment process. Finally, HybridMimic permits reduced joint stiffness, yielding compliant behavior for safer hardware operation.

Although HybridMimic effectively regulates ground reaction forces for stability and performance, the current feedforward formulation does not explicitly optimize swing-leg trajectories, which are critical for precise foot placement. Future work will explore integrating model-based swing-leg control, such as task-space control, into the existing architecture to enhance the tracking accuracy during leg-swing phases of highly dynamic maneuvers.

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